

REGULAR EXPRESSION

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AP/CSE

LEARNING OBJECTIVE

- To construct finite automata for any given pattern and find its equivalent regular expressions
 - To understand what is Regular Expression

INTRODUCTION

- Regular expressions describe regular languages
- ie the language accepted by a finite automata are easily described by regular expression.
- Many programming languages provide regular expression capabilities,
 - Built-in → Perl, JavaScript, Ruby, AWK, Tcl,
 - Standard library → .NET, Java, Python C++
- REs are widely supported in programming languages, text processing programs (particular lexers, lex, yacc), advanced text editors

INTRODUCTION

- Let Σ be a finite set of symbols.
- Let L_1, L_2 be set of strings in Σ^* .
- The concatenation of L_1 and L_2 denoted by $L_1 L_2$ is the set of all strings of the form xy , where $x \in L_1$ and $y \in L_2$.
- $L^0 = \{\epsilon\}$
- $L^i = LL^{i-1}$ for $i \geq 1$.

$$\begin{aligned} L_1 \cup L_2 &= \{x, x \in L_1, \text{ or } x \in L_2\} \\ L_1 \cdot L_2 &= \{xy \mid x \in L_1, y \in L_2\} \\ \underline{L_1^*} &= \bigcup_{i=0}^{\infty} L^i \end{aligned}$$

INTRODUCTION

- Kleene Closure

$$L^* = \bigcup_{i=0}^{\infty} L^i = L^0 \cup L^1 \cup L^2 \cup \dots$$

- Positive Closure

$$L^+ = \bigcup_{i=1}^{\infty} L^i = L^1 \cup L^2 \cup \dots$$

EXAMPLE

Let $L_1 = \{10, 01\}$, $L_2 = \{11, 00\}$

Then $L_1 L_2 = \{1011, 1000, 0111, 0100\}$

$$L_1 \cup L_2 = \{00, 11, 10, 01\}$$

Let $L = \{10, 11\}$

Then $L^* = L_0 \cup L_1 \cup L_2 \cup \dots$

$= \{\epsilon\} \cup \{10, 11\} \cup \{1011, 1010, 1110, 1111\} \cup \dots$

$= \{\epsilon, 10, 11, 1011, 1010, 1110, 1111, \dots\}$

OPERATORS OF RE

$$* \rightarrow L^*$$

$$\cdot \rightarrow L_1 \cdot L_2, L_1 L_2$$

$$/ \rightarrow \underline{L_1 \cup L_2}$$

Union / or $\pm$
 Con $\cdot$ $\underline{L_1 L_2}$
 Clo $*$

$$\begin{aligned} \underline{r_1} \quad \underline{r_2} &\longrightarrow RE \\ \underline{r_1} / \underline{r_2} \quad (or) \quad \underline{r_1 \pm r_2} \\ \underline{r_1} \cdot \underline{r_2} \quad (or) \quad \underline{r_1 r_2} \\ \underline{r_1}^* \end{aligned}$$

DEFINITION OF REGULAR EXPRESSION

- Let Σ be an alphabet. The regular expressions over Σ and the sets that they denote are defined recursively as follows:
 - ϕ is a regular expression and denotes the empty set $\{\}$. ϕ
 - ϵ is a regular expression and denotes the set $\{\epsilon\}$. ϵ
 - For each $a \in \Sigma$, ' a ' is a regular expression and denotes the set $\{a\}$.
 - If r and s are regular expressions denoting the languages R and S respectively then $(r + s)$, (rs) , $(r)^*$ are regular expressions that denotes the sets $R \cup S$, RS and R^* respectively.

$\Sigma = \{a, b\}$
 $\frac{a}{RE}$ b $/$ r/s $R \cup S \rightarrow \cup$
 $\cdot$ $r.s$ RS
 $*$ r^* R^*

PRECEDENCE OF RE OPERATORS

* → higher precedence

.

/ → Lower precedence

EXAMPLE

- $(0/1)^* = \{\epsilon, 0, 1, 00, 01, 10, 11, \dots\} = (0+1)^*$ (i.e.) all strings of 0 and 1
- 01^* = $\{0, 01, 011, 0111, \dots\}$
- 0^* = $\{\epsilon, 0, 00, 000, \dots\}$
- $1(1)^* = \{1, 11, 111, 1111, \dots\} = 1^+$

Handwritten notes:

$$(0/1)^* = \{0, 1\}$$

$$0/1^* = \{0, 1, 11, \dots\}$$

$$(0/1)^* = \{\epsilon, 0, 1, 00, 11, \dots\}$$

Handwritten notes:

$$0 = \{0\}$$

$$1^* = \{\epsilon, 1, 11, 111, \dots\}$$

$$\underline{01^*} = \{0, 01, 011, 0111, \dots\}$$

Handwritten notes:

$$1(1)^* = \{1, 11, 111, \dots\}$$

$$1 = \{1\}$$

$$11^* = \{1, 11, 111, \dots\}$$

EXAMPLE

$$L = \{0, 1\} \quad RE? \quad (0/1)^*$$

$$L = \{0^n \mid n \text{ is } 5\}$$

$$\underline{L = \{00000\}}$$

$$\underline{0} \quad \underline{L = \{0\}}$$

$$RE = ?$$

$$\underline{\underline{00000}}$$

$$L = \{1^m \mid m \text{ is divisible by } 2 \text{ or } 7\}$$

$$L = \{\epsilon, \underline{1}, \underline{11}, \underline{1111}, \underline{111111}, \dots\}$$

$$RE = ? \quad (\underline{11})^* \mid (\underline{111111})^*$$

EXAMPLE

- The set of all strings over $\{0,1\}$ with three consecutive 0's.

L=? $\{000, 1000, \dots\}$

RE=? $(0/1)^*000(0/1)^*$

- The set of all strings over $\{0,1\}$ beginning with 00.

L=? $L = \{00, 00\dots\}$

RE=? $00(0/1)^*$

EXAMPLE

- $\{1, 11, 111, \dots\}$

L=?

RE=?

- The set of all strings over $\{0, 1\}$ which has at most two 0's.

L=?

RE=?

$(0/1)^*$ $1^*01^*01^*$ + 1^*01^* + 1^*

IDENTITIES FOR REGULAR EXPRESSIONS

$$I1 \quad \phi + R = R$$

$$I2 \quad \phi R = R\phi = \phi$$

$$I3 \quad \lambda R = R\lambda = R$$

$$I4 \quad \lambda^* = \underline{\lambda}$$

$$I5 \quad R + R = R$$

$$I6 \quad R^*R^* = R^*$$

$$I7 \quad RR^* = R^*R$$

$$I8 \quad (R^*)^* = R^*$$

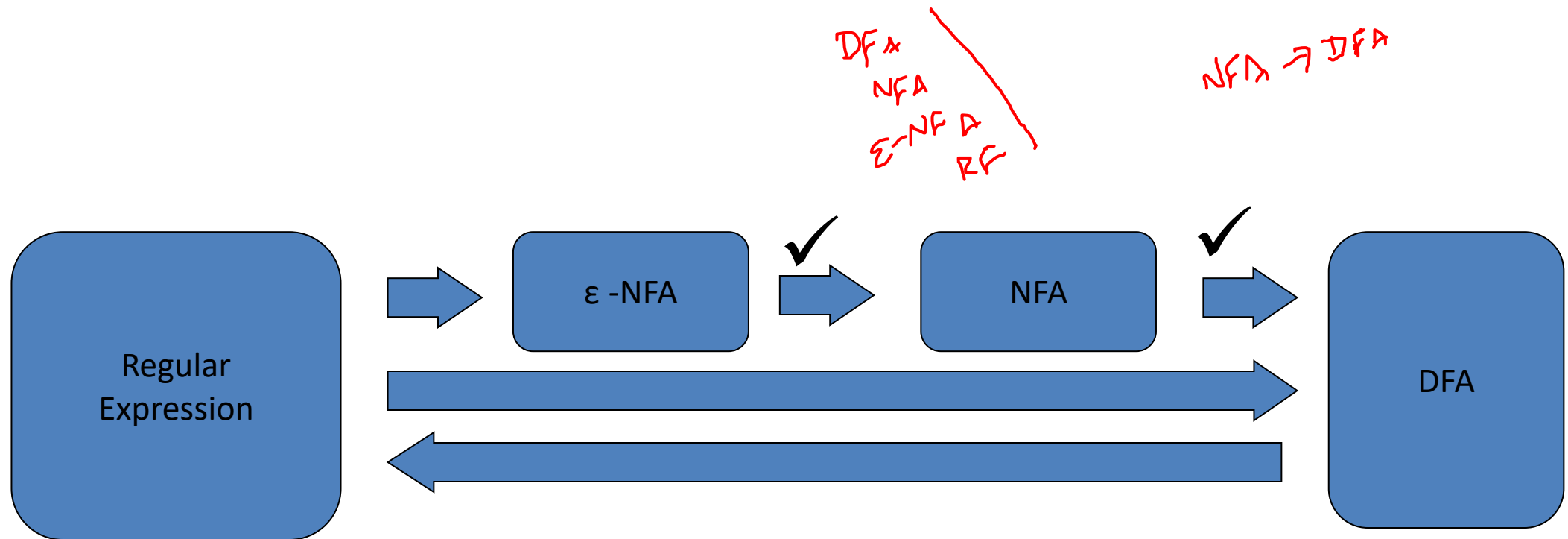
$$I9 \quad \lambda + RR^* = R^* = \lambda + R^*R$$

$$I10 \quad (PQ)^*P = P(QP)^*$$

$$I11 \quad (P + Q)^* = (P^*Q^*)^* = (P^* + Q^*)^*$$

$$I12 \quad (P + Q)R = PR + QR \text{ and } R(P + Q) = RP + RQ$$

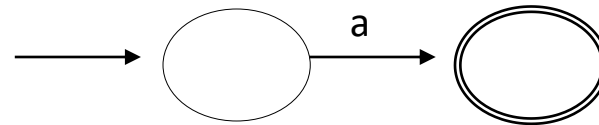
ROAD MAP



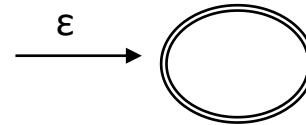
THOMPSON'S CONSTRUCTION

- Basis

$R=a$

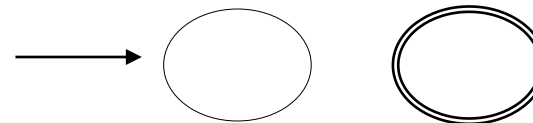


$R=\epsilon$



$R=\phi$

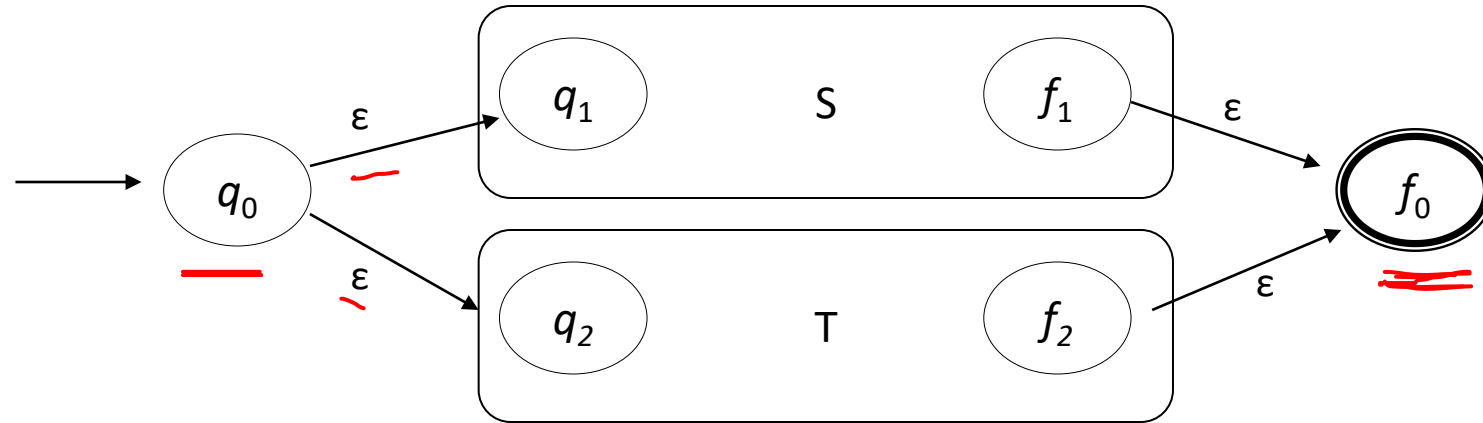
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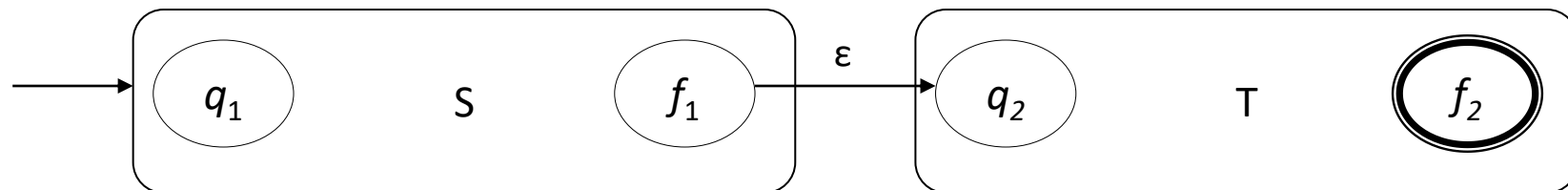
THOMPSON'S CONSTRUCTION

4 i) ST
S/T

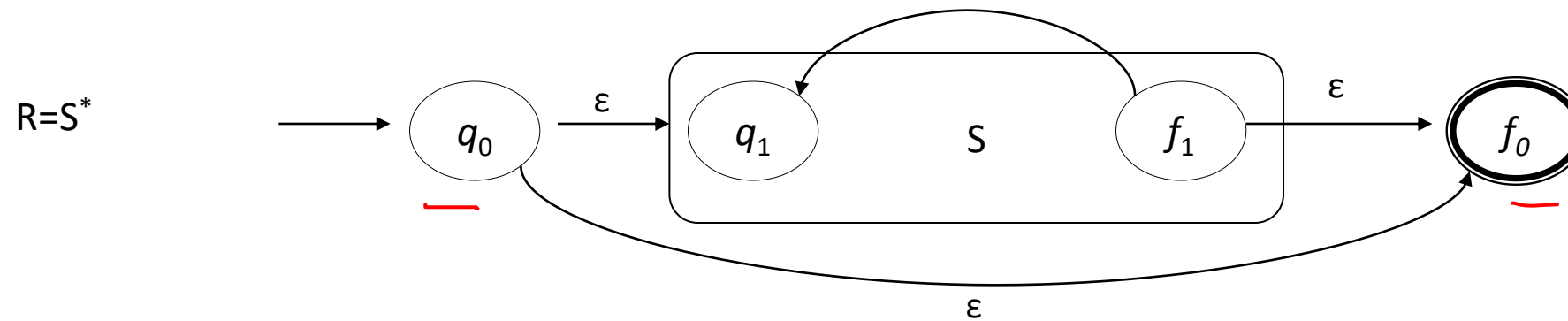
$R = S + T$



$R = \underline{ST}$

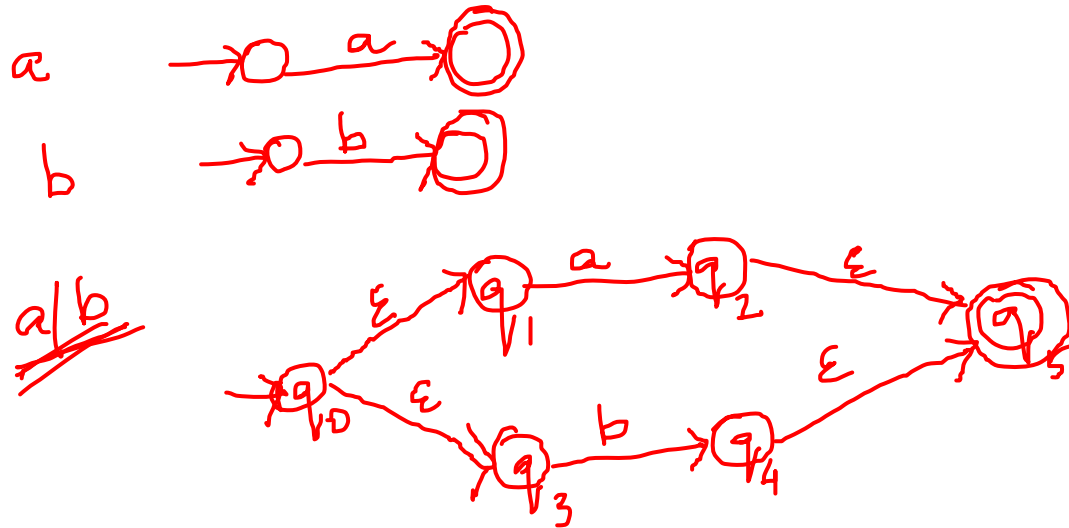


THOMPSON'S CONSTRUCTION

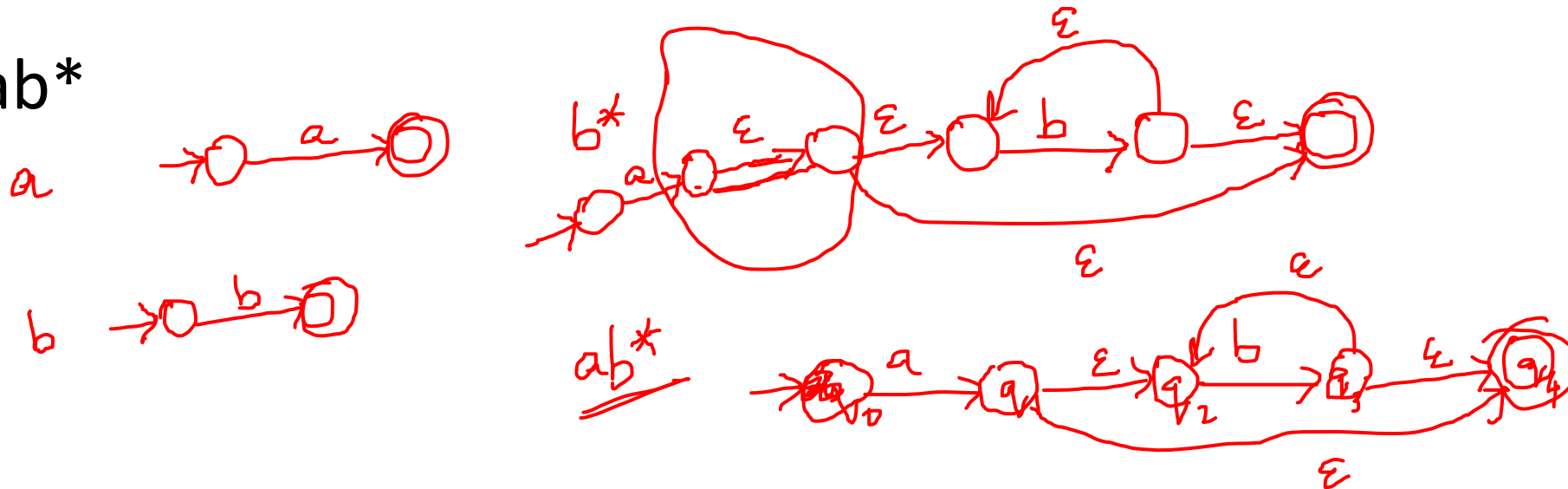


THOMPSON'S CONSTRUCTION

- a/b
a, b.

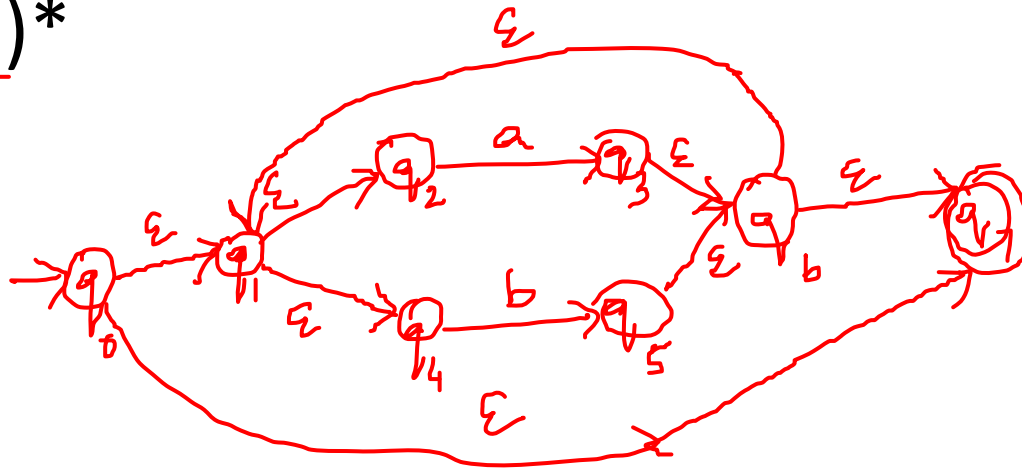


- ab^*



THOMPSON'S CONSTRUCTION

- $(a/b)^*$



$L = \{\epsilon, a, b, ab, ba, bba, \dots\}$

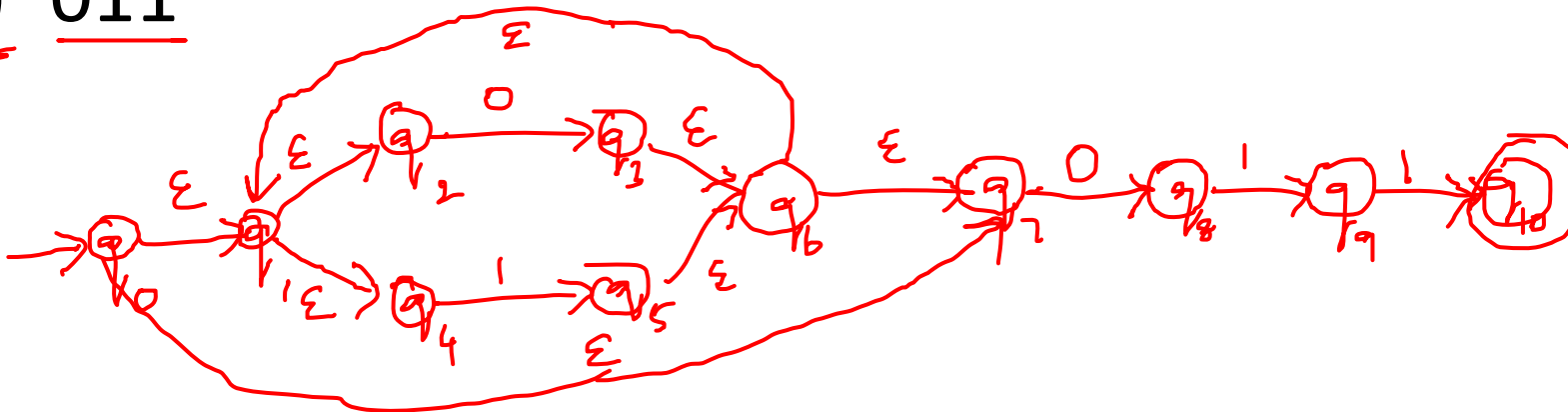
~~$\{ \epsilon, a, aa \}$~~

$(0/1)^* (0/1)^*$

$10(01/0)^*$

$(11/00)(0/1)^*$

- $(0/1)^*011$



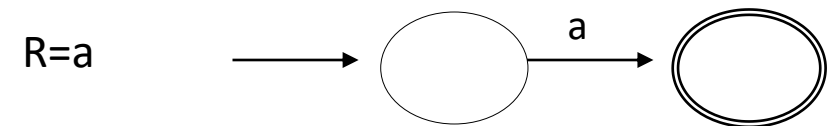
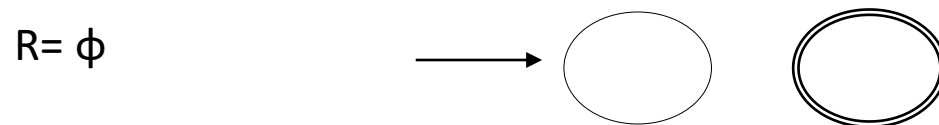
THEOREM

For every regular expression r there exists a NFA with ε -transitions that accepts $L(r)$

- Proof
 - **Basis step (Zero operators)**

Suppose r is ε , ϕ or a for some $a \in \Sigma$.

Then the equivalent NFA's are:



INDUCTION CASE I

- $r = r_1 + r_2$
- $M_1 = (Q_1, \Sigma_1, \delta_1, q_1, \{f_1\})$ $L(M_1) = L(r_1)$
- $M_2 = (Q_2, \Sigma_2, \delta_2, q_2, \{f_2\})$ $L(M_2) = L(r_2)$.
- Assume Q_1 and Q_2 are disjoint.
- Let q_0, f_0 be a new initial and final state respectively.

CASE I

- $M = (Q_1 \cup Q_2 \cup \{q_0, f_0\}, \Sigma_1 \cup \Sigma_2, \delta, q_0, \{f_0\})$ where δ is defined by
$$\delta(q_0, \epsilon) = \{q_1, q_2\}$$
$$\delta(q, a) = \delta_1(q, a) \quad \text{if } q \in Q_1 - \{f_1\}, a \in \Sigma_1 \cup \{\epsilon\}$$
$$\delta(q, a) = \delta_2(q, a) \quad \text{if } q \in Q_2 - \{f_2\}, a \in \Sigma_2 \cup \{\epsilon\}$$
$$\delta_1(f_1, \epsilon) = \delta_2(f_2, \epsilon) = \{f_0\}$$

- $L(M) = L(M_1) \cup L(M_2)$

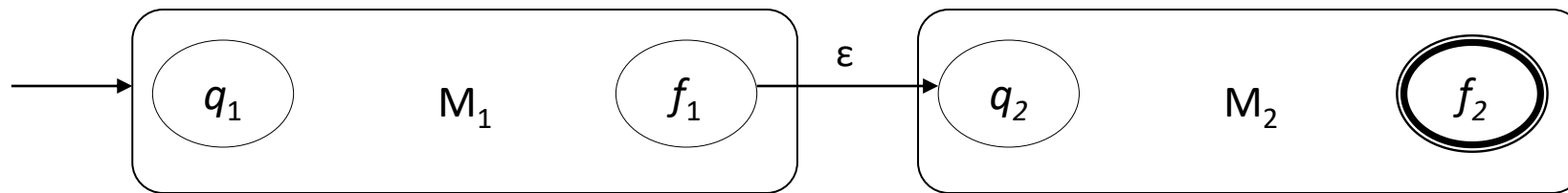


CASE II

- $r = r_1 \cdot r_2$
- $M_1 = (Q_1, \Sigma_1, \delta_1, q_1, \{f_1\})$ $L(M_1) = L(r_1)$
- $M_2 = (Q_2, \Sigma_2, \delta_2, q_2, \{f_2\})$ $L(M_2) = L(r_2)$
- $M = (Q_1 \cup Q_2, \Sigma_1 \cup \Sigma_2, \delta, \{q_1\}, \{f_2\})$, where δ is given by:
 $\delta(q, a) = \delta_1(q, a)$ for q in $Q_1 - \{f_1\}$ and a in $\Sigma_1 \cup \{\epsilon\}$
 $\delta(f_1, \epsilon) = \{q_2\}$
 $\delta(q, a) = \delta_2(q, a)$ for q in Q_2 and a in $\Sigma_2 \cup \{\epsilon\}$

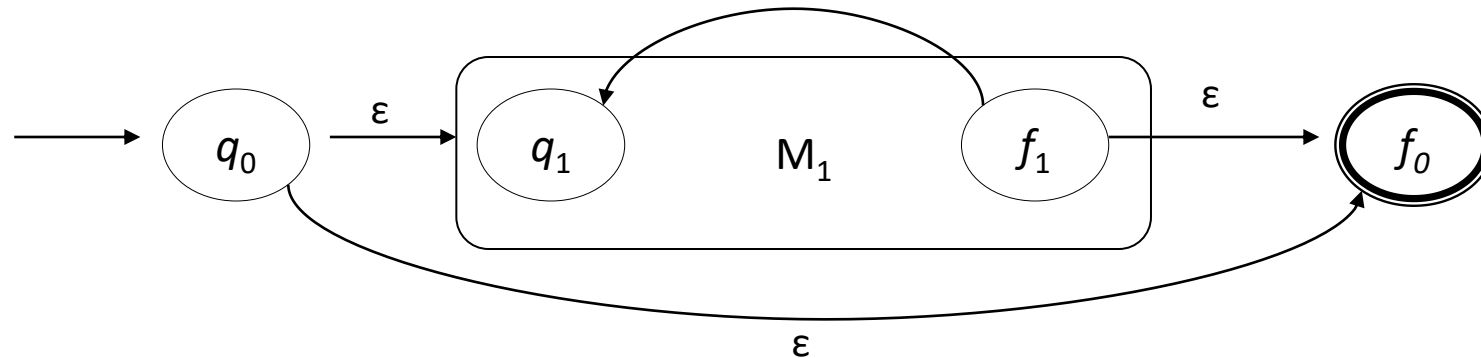
CASE II

- $L(M) = \{xy \mid x \text{ is in } L(M_1) \text{ and } y \text{ is in } L(M_2)\}$
- $L(M) = L(M_1) \cdot L(M_2).$

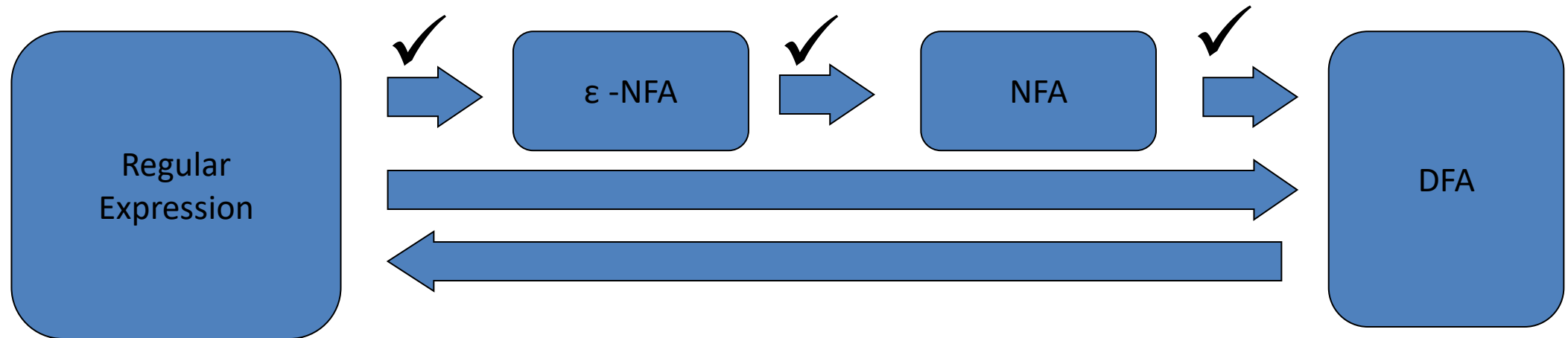


CASE III

- $r = r_1^*$
- $M_1 = (Q_1, \Sigma_1, \delta_1, q_1, \{f_1\})$ $L(M_1) = r_1$
- $M = (Q_1 \cup \{q_0, f_0\}, \Sigma_1, \delta, q_0, \{f_0\})$, where δ is given by:
 $\delta(q_0, \varepsilon) = \delta(f_1, \varepsilon) = \{q_1, f_0\}$
 $\delta(q, a) = \delta_1(q, a)$ for q in $Q_1 - \{f_1\}$ and a in $\Sigma_1 \cup \{\varepsilon\}$



ROAD MAP



CONVERSION OF E-NFA TO DFA

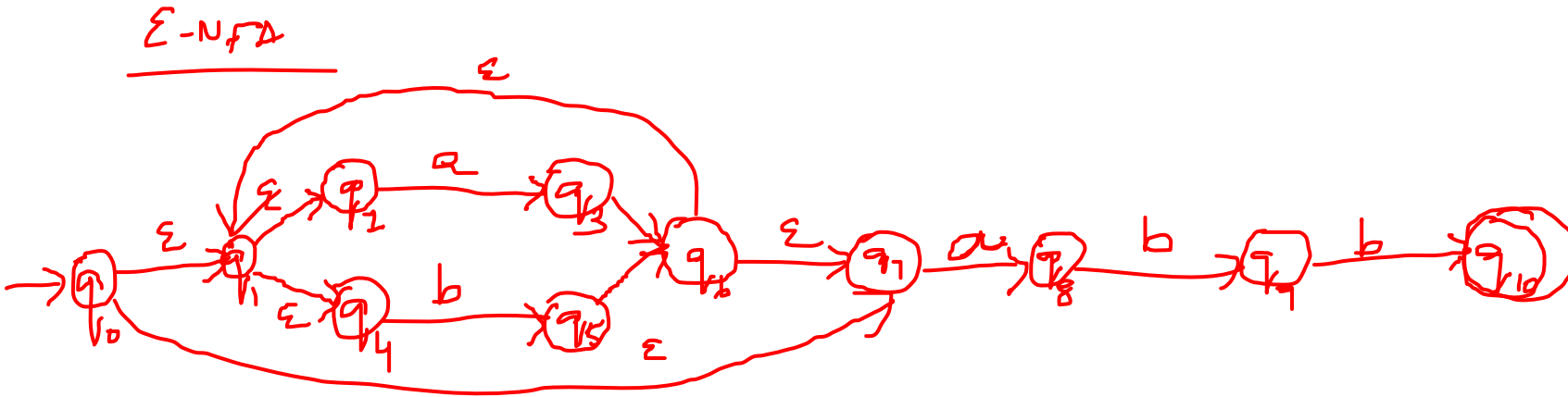
1. Find the ϵ -CLOSURE of the state q_0 from the constructed ϵ -NFA (i.e) from state q_0 , ϵ transition to other states are identified as well as ϵ transitions from other states are also identified and combined as one set (new state).

CONVERSION OF E-NFA TO DFA

2. Perform the following steps until there are no more new states as been constructed.
 - i. Find the transition of the given regular expression symbols over Σ from the new state (i.e) move (new state, symbol)
 - ii. Find the ε -CLOSURE of move (new state, symbol).

EXAMPLE

$(a/b)^*abb$

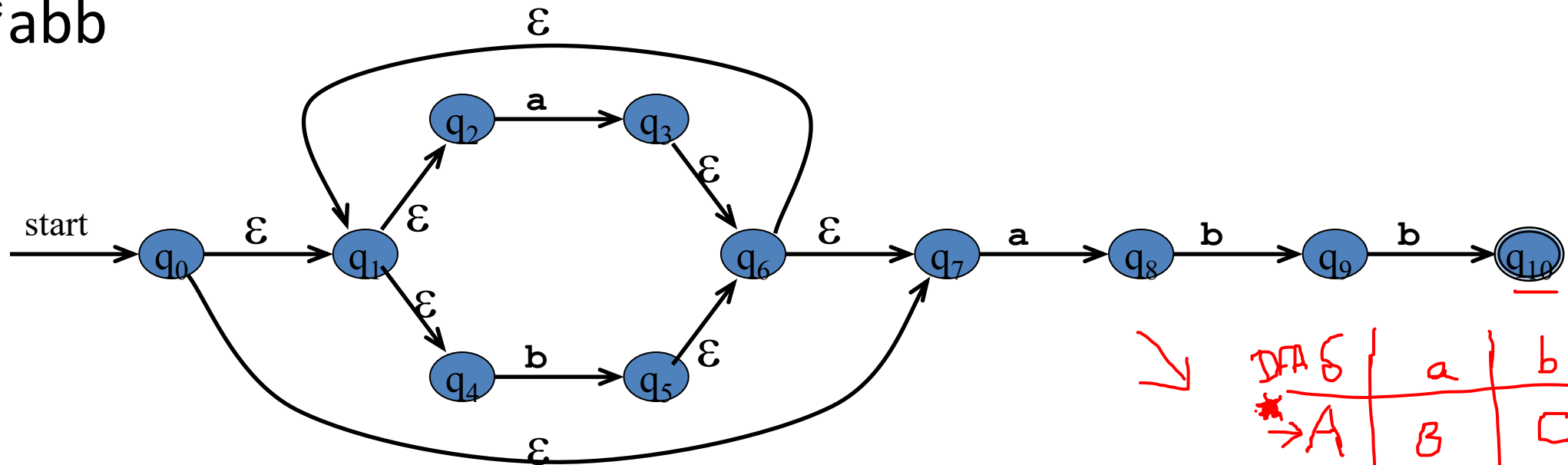


ϵ -NFA $\rightarrow$ DFA subset construction

ϵ -NFA $\rightarrow$ NFA $\rightarrow$ DFA

EXAMPLE

$(a/b)^*abb$



$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2, q_4, q_7\} = A$$

$$\text{Mov}(A, a) = \{q_3, q_6\} / \epsilon\text{-closure}(\text{Mov}(A, a)) = \{q_3, q_6, q_2, q_1, q_7, q_4\} = B$$

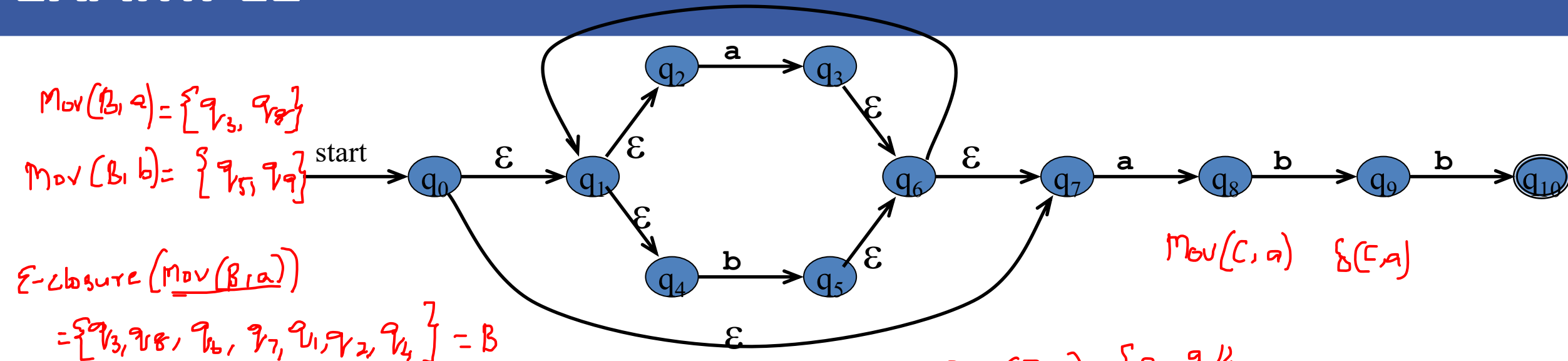
$$\text{Mov}(A, b) = \{q_5\} / \epsilon\text{-closure}(\text{Mov}(A, b)) = \{q_5, q_6, q_1, q_7, q_2, q_4\} = C$$

↘

DFA δ	a	b
* $\rightarrow A$	B	C
B	B	D
C	B	C

	a	b
D	B	E
* E	B	C

EXAMPLE



$$Mov(B, a) = \{q_3, q_8\}$$

$$Mov(B, b) = \{q_5, q_9\}$$

$$\epsilon\text{-closure}(Mov(B, a)) = \{q_3, q_8, q_6, q_7, q_1, q_2, q_4\} = B$$

$$\epsilon\text{-closure}(Mov(B, b)) = \{q_5, q_9, q_6, q_7, q_1, q_2, q_4\} = D$$

$$\begin{array}{l} Mov(C, a) = \{q_3, q_8\} \\ Mov(C, b) = \{q_5\} \end{array} \quad \left| \begin{array}{l} \epsilon\text{-closure}(Mov(C, a)) = B \\ \epsilon\text{-closure}(Mov(C, b)) = C \end{array} \right.$$

$$Mov(D, a) = \{q_3, q_8\} \quad \left| \quad \epsilon\text{-closure}(Mov(D, a)) = B \right.$$

$$Mov(D, b) = \{q_5, q_{10}\} \quad \left| \quad \epsilon\text{-closure}(Mov(D, b)) = \{q_5, q_{10}, q_6, q_7, q_1, q_2, q_4\} = E^*$$

$$Mov(C, a) \quad \delta(E, a)$$

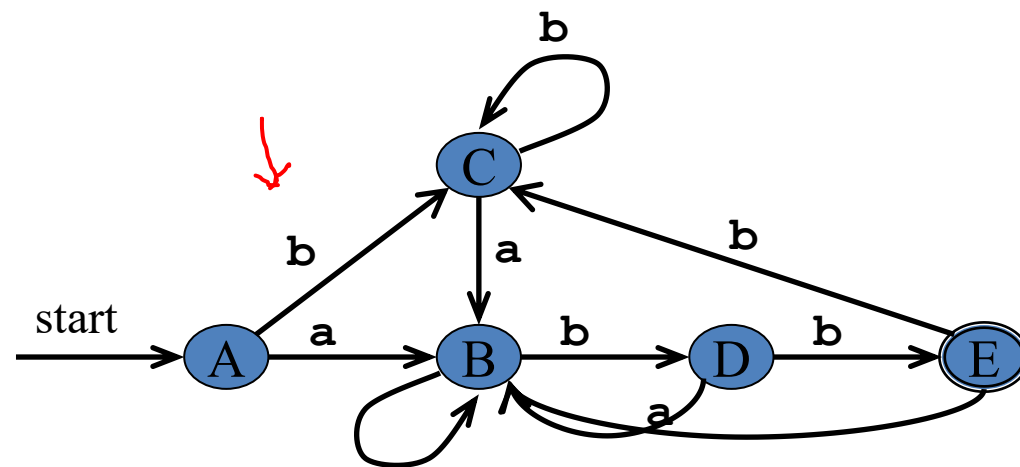
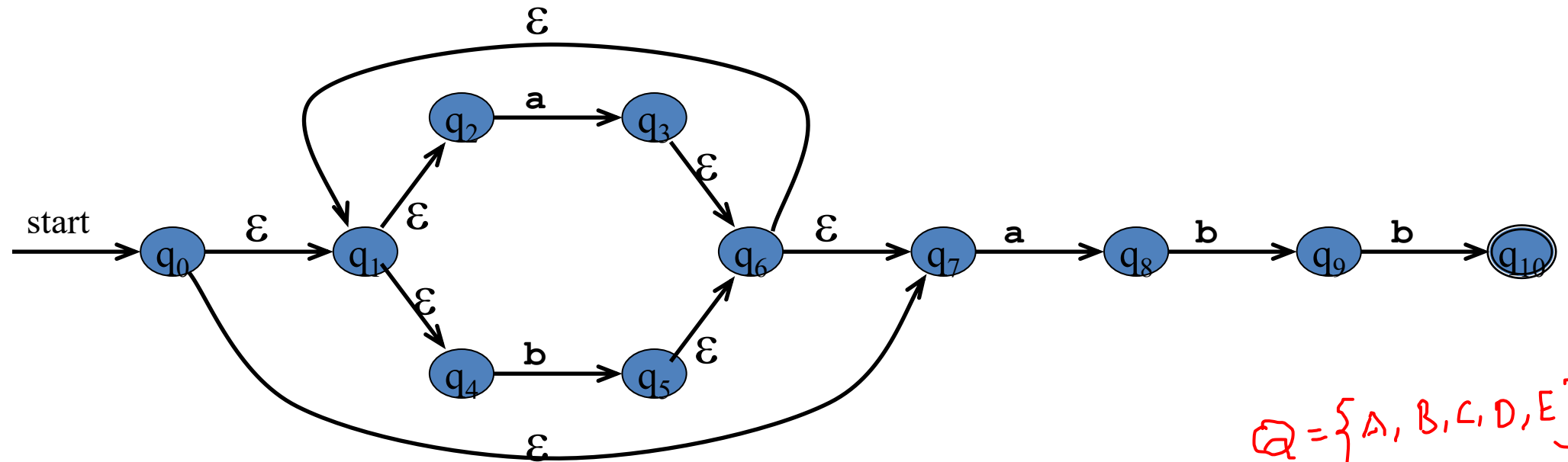
$$Mov(E, a) = \{q_3, q_8\}$$

$$Mov(E, b) = \{q_5\}$$

$$\epsilon\text{-closure}(Mov(E, a)) = B$$

$$\epsilon\text{-closure}(Mov(E, b)) = C$$

EXAMPLE



~~Dstates~~

~~$A = \{0, 1, 2, 4, 7\}$~~

~~$B = \{1, 2, 3, 4, 6, 7, 8\}$~~

~~$C = \{1, 2, 4, 5, 6, 7\}$~~

~~$D = \{1, 2, 4, 5, 6, 7, 9\}$~~

~~$E = \{1, 2, 4, 5, 6, 7, 10\}$~~

$Q = \{A, B, C, D, E\}$

$\Sigma = \{a, b\}$

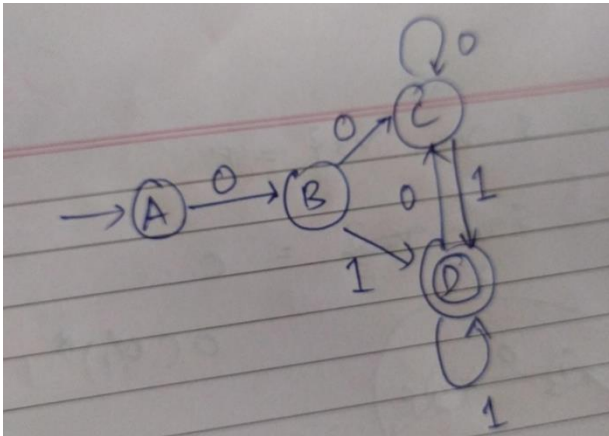
$A \rightarrow \text{TS}$

$F = \{E\}$

δ

EXAMPLE

- $0(0/1)^*1$



Handwritten notes showing the construction of the NFA and the calculation of the ϵ -closure and move functions for the regular expression $0(0/1)^*1$.

NFA Diagram: The diagram shows states $v_0, v_1, v_2, v_3, v_4, v_5, v_6, v_7$. v_0 is the start state and v_7 is the final state. Transitions include $v_0 \xrightarrow{0} v_1$, $v_1 \xrightarrow{0} v_2$, $v_1 \xrightarrow{1} v_6$, $v_2 \xrightarrow{0} v_3$, $v_3 \xrightarrow{0} v_4$, $v_4 \xrightarrow{0} v_5$, $v_4 \xrightarrow{1} v_6$, $v_5 \xrightarrow{0} v_4$, $v_6 \xrightarrow{1} v_7$, and $v_7 \xrightarrow{1} v_6$. There is also a self-loop on v_4 for input 0.

ϵ -closure and move Calculations:

$\epsilon\text{-closure}(v_0) = \{v_0\} = A$

$\text{move}(v_0, 0) = \{v_1\}$

$\epsilon\text{-closure}(\text{move}(v_0, 0)) = \{v_1, v_2, v_3, v_4, v_5\} = B$

$\text{move}(v_0, 1) = \emptyset$

$\epsilon\text{-closure}(\text{move}(v_0, 1)) = \emptyset$

$\text{move}(B, 0) = \{v_4\}$

$\epsilon\text{-closure}(\text{move}(B, 0)) = \{v_4, v_5, v_6, v_7\} = C$

$\text{move}(B, 1) = \{v_6, v_7\}$

$\epsilon\text{-closure}(\text{move}(B, 1)) = \{v_6, v_7, v_5, v_4, v_3, v_2, v_1\} = D$

$\text{move}(C, 0) = \{v_4\} \Rightarrow \epsilon\text{-closure}(\text{move}(C, 0)) = D$

$\text{move}(C, 1) = \{v_6, v_7\} \Rightarrow \epsilon\text{-closure}(\text{move}(C, 1)) = D$

$\text{move}(D, 0) = \{v_4\} \Rightarrow \epsilon\text{-closure}(\text{move}(D, 0)) = C$

$\text{move}(D, 1) = \{v_6, v_7\} \Rightarrow \epsilon\text{-closure}(\text{move}(D, 1)) = D$

SUMMARY

- Definition of RE
- Precedence, identities, properties of RE.
- Thomson's construction to convert RE to NFA and then to DFA

TEST YOUR KNOWLEDGE

- Which of the following does not represents the given language?

Language: $\{0,01\}$

- a) $0+01$
- b) $\{0\} \cup \{01\}$
- c) $\{0\} \cup \{0\}\{1\}$
- d) $\{0\} \wedge \{01\}$

TEST YOUR KNOWLEDGE

- Regular Expression R and the language it describes can be represented as:
 - a) $R, R(L)$
 - b) $L(R), R(L)$
 - c) $R, L(R)$
 - d) All of the mentioned

LEARNING OUTCOME

On successful completion of this topic, the student will be able to:

- Understand the concepts of RE (K3)

REFERENCE

- Hopcroft J.E., Motwani R. and Ullman J.D, “Introduction to Automata Theory, Languages and Computations”, Second Edition, Pearson Education, 2008